AI-Powered Cybersecurity Incident Response Assistant

Overview:
Cybersecurity teams continuously monitor network traffic for suspicious activity such as brute-force attacks, denial-of-service attacks, web attacks, botnet activity, infiltration, and port scanning. Security analysts need to investigate alerts, identify similar historical incidents, classify attack types, determine the severity of suspicious activity, and prioritize incidents for investigation. Manually analyzing large volumes of network-flow records can be time-consuming and difficult.

Example Professional Questions:

- Has a similar network security incident occurred previously, and what type of attack was involved?
- What network traffic characteristics are associated with a particular attack type?
- Which suspicious network activities should be prioritized for investigation?

Instead of manually searching through large volumes of network traffic and security-event records, build an AI-powered assistant that can retrieve similar incidents, classify attack patterns, identify suspicious activity, and support security analysts during incident investigation.

Build a semantic retrieval and AI-assisted cybersecurity incident analysis system that can help security teams investigate, classify, and prioritize network security incidents.

Requirement 1 – Basic

- Build a Basic RAG system for historical cybersecurity incident similarity search
- Implement hybrid search using keyword and semantic retrieval
- Build a cybersecurity incident classification agent
- Implement metadata filtering based on attack type, protocol, source/destination information, ports, and timestamp
- Generate basic incident investigation and response recommendations
- Implement input validation guardrails
- Implement incident prioritization based on attack characteristics and traffic indicators
- Expose the core functionality through API endpoints.

Requirement 2 – Advanced

- Implement DeepEval for evaluating incident-analysis and recommendation quality
- Build an attack-type and intrusion-risk prediction model
- Implement LLM-as-Judge for validating incident analysis and recommendations
- Build a multi-agent cybersecurity investigation workflow for detection, analysis, risk assessment, and response recommendation

- Implement anomaly and suspicious-traffic risk prediction
- Perform explainable analysis of network-traffic features and their relationship to attack types
- Implement a feedback loop for continuous improvement
- Build a simple front-end interface.

Deliverables:

1. **Architecture Diagram**
2. **Design**
3. **Full Executable Code (Microservice)**
4. **Panel Presentation (10 minutes)**

Dataset Name:

CIC-IDS2017 – Intrusion Detection Evaluation Dataset

Primary Link:

CIC-IDS2017 Dataset – Canadian Institute for Cybersecurity

Alternative Link:

CIC-IDS2017 – Kaggle (clean Parquet version)

Format:

CSV / PCAP

Key Fields:

Flow ID, Source IP, Source Port, Destination IP, Destination Port, Protocol, Timestamp, Flow Duration, Total Fwd Packets, Total Backward Packets, Total Length of Fwd Packets, Total Length of Bwd Packets, Flow Bytes/s, Flow Packets/s, Flow IAT Mean, Packet Length Mean, SYN Flag Count, RST Flag Count, ACK Flag Count, Average Packet Size, Active Mean, Idle Mean, Label

Reason:

CIC-IDS2017 contains labeled benign and malicious network traffic and includes a wide range of network-flow characteristics. The dataset covers attack categories such as FTP/SSH brute force, DoS, DDoS, Heartbleed, web attacks, infiltration, botnet activity, and port scanning. Its network-flow features, timestamps, IP/port information, and attack labels support semantic incident retrieval, hybrid search, attack classification, metadata filtering, incident prioritization, intrusion-risk prediction, anomaly analysis, explainability, and AI-assisted cybersecurity investigation.